

TC74HC165P/F

C²MOS DIGITAL INTEGRATED CIRCUIT

PRELIMINARY

TC74HC165P/F 8-BIT SHIFT REGISTER (P-IN, S-OUT)

The TC74HC165 is a high speed CMOS 8-BIT PARALLEL/SERIAL-IN SERIAL-OUT SHIFT REGISTER fabricated with silicon gate C²MOS technology.

It achieves the high speed operation similar to equivalent LSTTL while maintaining the CMOS low power dissipation.

It consists of parallel-in or serial-in, serial-out 8-bit shift register with gated clock input.

When the SHIFT/LOAD input is held high, the serial data input is enabled and the eight flip-flops perform serial shifting with each clock pulse.

When the SHIFT/LOAD input is held low, the parallel data is loaded asynchronously into the register.

Clocking is accomplished on the positive going transition of the clock pulse.

The CLOCK-INHIBIT input should be changed to the high level only while the CLOCK input is held high.

All inputs are equipped with protection circuits against static discharge or transient excess voltage.

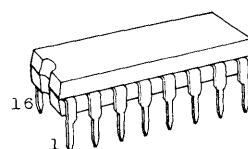
FEATURES:

- High Speed $f_{MAX}=48\text{MHz(Typ.)}$ at $V_{CC}=5\text{V}$
- Low Power Dissipation $I_{CC}=4\mu\text{A(Max.)}$ at $T_a=25^\circ\text{C}$
- High Noise Immunity $V_{NIH}=V_{NIL}=28\% V_{CC}(\text{Min.})$
- Output Drive Capability 10 LSTTL Loads
- Symmetrical Output Impedance $|I_{OH}|=I_{OL}=4\text{mA}(\text{Min.})$
- Balanced Propagation Delay $t_{PLH}=t_{PHL}$
- Wide Operating Voltage Range $V_{CC}(\text{Opr.})=2\text{V} \sim 6\text{V}$
- Pin and Function Compatible with 74LS165

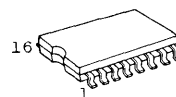
ABSOLUTE MAXIMUM RATINGS

PARAMETER	SYMBOL	VALUE	UNIT
Supply Voltage Range	V_{CC}	$-0.5 \sim 7$	V
DC Input Voltage	V_{IN}	$-0.5 \sim V_{CC}+0.5$	V
DC Output Voltage	V_{OUT}	$-0.5 \sim V_{CC}+0.5$	V
Input Diode Current	I_{IK}	± 20	mA
Output Diode Current	I_{OK}	± 20	mA
DC Output Current	I_{OUT}	± 25	mA
DC V_{CC} /Ground Current	I_{CC}	± 50	mA
Power Dissipation	P_D	500(DIP)* 180(MFP)	mW
Storage Temperature	T_{stg}	$-65 \sim 150$	$^\circ\text{C}$
Lead Temperature 10sec	T_L	300	$^\circ\text{C}$

* 500mW in the range of $T_a=-40^\circ\text{C} \sim 65^\circ\text{C}$ and from $T_a=65^\circ\text{C}$ up to 85°C derating factor of $-10\text{mW}/^\circ\text{C}$ shall be applied until 300mW.

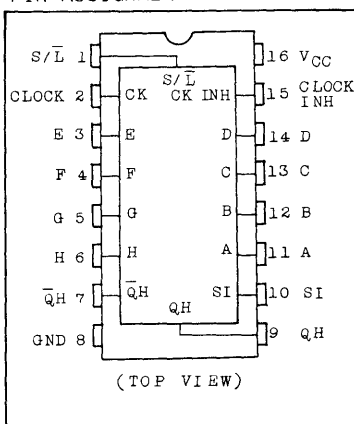


DIP16(3D16A-P)



MFP16(F16GC-P)

PIN ASSIGNMENT



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TRUTH TABLE

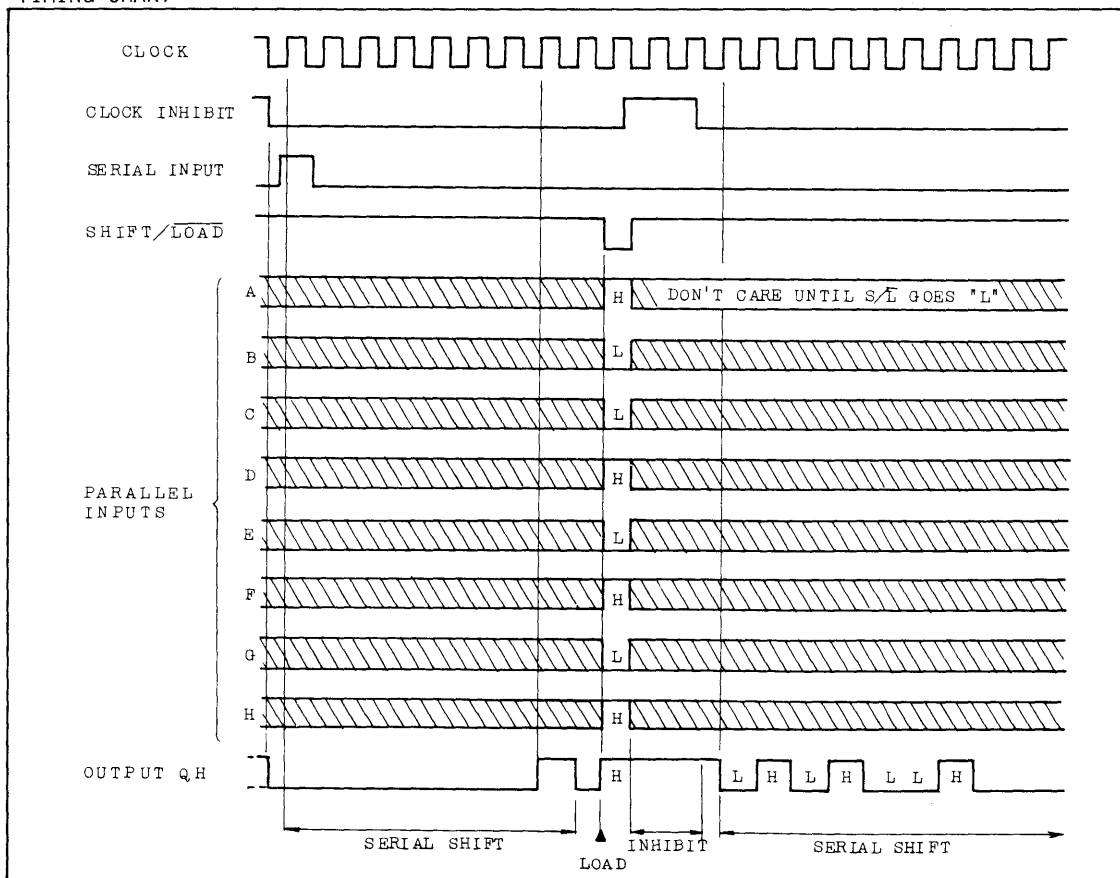
SHIFT/ LOAD	CLOCK INH	INPUTS			INTERNAL OUTPUTS		OUTPUT QH
		CLOCK	SERIAL IN	PARALLEL A H	QA	QB	
L	X	X	X	a h	a	b	h
H	L		H	X	H	QAn	QGn
H	L		L	X	L	QAn	QGn
H		L	H	X	H	QAn	QGn
H		L	L	X	L	QAn	QGn
H	X	H	X	X	No change		
H	H	X	X	X	No change		

X : DON'T CARE

a h : THE LEVEL OF STEADY INPUT VOLTAGE AT INPUTS A THROUGH H RESPECTIVELY

QAn...QGn: THE LEVEL OF QA~QG, RESPECTIVELY, BEFORE THE MOST-RECENT POSITIVE TRANSITION OF THE CLOCK.

TIMING CHART

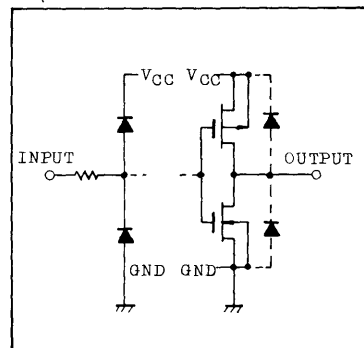


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RECOMMENDED OPERATING CONDITIONS

PARAMETER	SYMBOL	LIMIT	UNIT
Supply Voltage	V_{CC}	2 ~ 6	V
Input Voltage	V_{IN}	0 ~ V_{CC}	V
Output Voltage	V_{OUT}	0 ~ V_{CC}	V
Operating Temperature	T_{opr}	-40 ~ 85	°C
Input Rise and Fall Time	t_r, t_f	0 ~ 1000 ($V_{CC}=2.0V$) 0 ~ 500 ($V_{CC}=4.5V$) 0 ~ 400 ($V_{CC}=6.0V$)	ns

INPUT and OUTPUT
EQUIVALENT CIRCUIT

DC ELECTRICAL CHARACTERISTICS

PARAMETER	SYMBOL	TEST CONDITION	V_{CC}	$T_a=25^{\circ}C$			$T_a=-40\sim 85^{\circ}C$		UNIT
				MIN.	TYP.	MAX.	MIN.	MAX.	
High-Level Input Voltage	V_{IH}		2.0	1.5	-	-	1.5	-	V
			4.5	3.15	-	-	3.15	-	
			6.0	4.2	-	-	4.2	-	
Low-Level Input Voltage	V_{IL}		2.0	-	-	0.5	-	0.5	V
			4.5	-	-	1.35	-	1.35	
			6.0	-	-	1.8	-	1.8	
High-Level Output Voltage	V_{OH}	$V_{IN}=V_{IH}$ or V_{IL}	$I_{OH}=-20\mu A$	2.0	1.9	2.0	-	1.9	V
				4.5	4.4	4.5	-	4.4	
				6.0	5.9	6.0	-	5.9	
		$I_{OH}=-4mA$ $I_{OH}=-5.2mA$		4.5	4.18	4.31	-	4.13	
				6.0	5.68	5.80	-	5.63	
				6.0	5.68	5.80	-	5.63	
Low-Level Output Voltage	V_{OL}	$V_{IN}=V_{IH}$ or V_{IL}	$I_{OL}=20\mu A$	2.0	-	0.0	0.1	-	V
				4.5	-	0.0	0.1	-	
				6.0	-	0.0	0.1	-	
		$I_{OL}=4mA$ $I_{OL}=5.2mA$		4.5	-	0.17	0.26	-	
				6.0	-	0.18	0.26	-	
				6.0	-	0.18	0.26	-	
Input Leakage Current	I_{IN}	$V_{IN}=V_{CC}$ or GND	6.0	-	-	± 0.1	-	± 1.0	μA
Quiescent Supply Current	I_{CC}	$V_{IN}=V_{CC}$ or GND	6.0	-	-	4.0	-	40.0	

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AC ELECTRICAL CHARACTERISTICS (C_L=50pF, INPUT t_r=t_f=6ns)

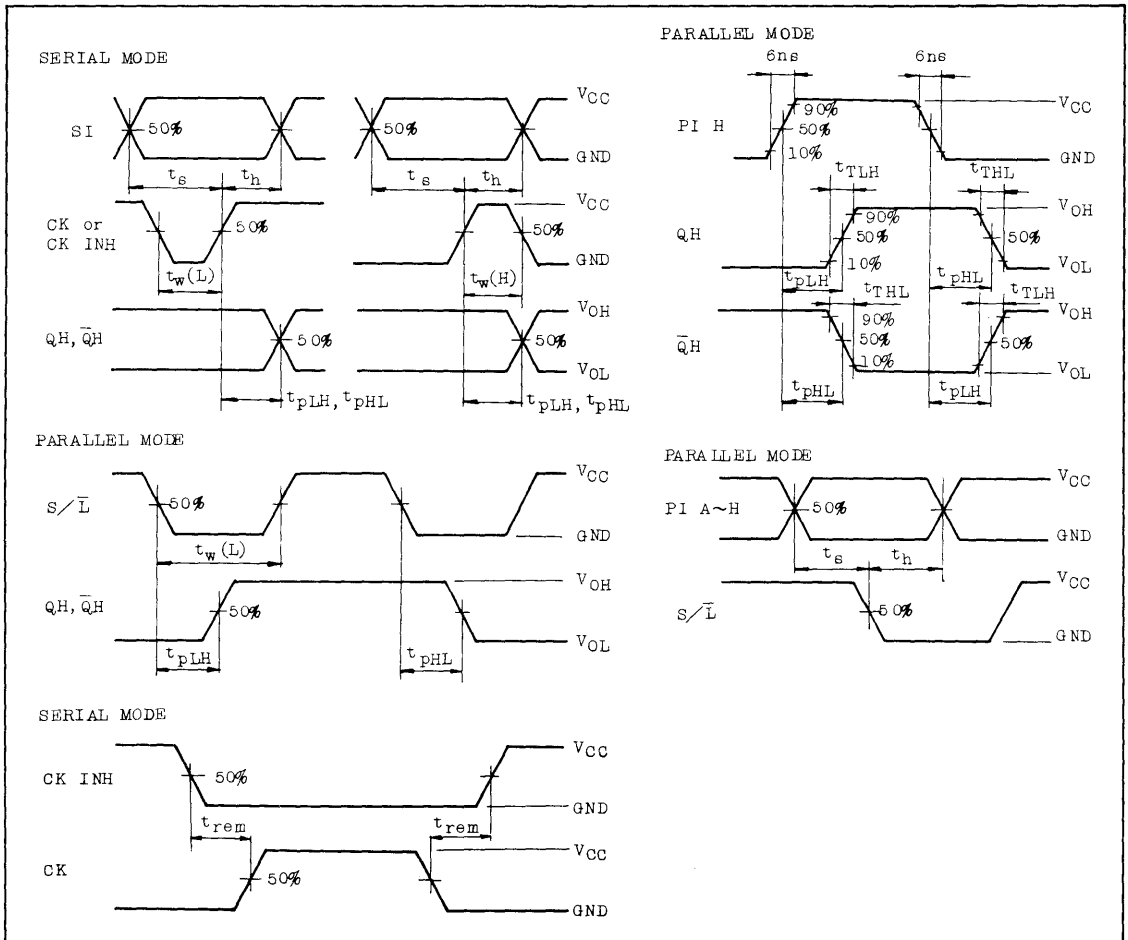
PARAMETER	SYMBOL	TEST CONDITION	Ta=25°C			Ta=-40~85°C		UNIT
			VCC	MIN.	TYP.	MAX.	MIN.	MAX.
Output Transition Time	t _{TLH} t _{THL}		2.0	-	30	75	-	95
			4.5	-	8	15	-	19
			6.0	-	7	13	-	16
Propagation Delay Time (CK, CK INH - QH, $\bar{Q}$ H)	t _{pLH} t _{pHL}		2.0	-	96	190	-	240
			4.5	-	24	38	-	48
			6.0	-	20	32	-	41
Propagation Delay Time (S/ $\bar{L}$ - QH, $\bar{Q}$ H)	t _{pLH} t _{pHL}		2.0	-	104	200	-	250
			4.5	-	26	40	-	50
			6.0	-	22	34	-	43
Propagation Delay Time (H - QH, $\bar{Q}$ H)	t _{pLH} t _{pHL}		2.0	-	92	180	-	225
			4.5	-	23	36	-	45
			6.0	-	20	31	-	38
Maximum Clock Frequency	f _{MAX}		2.0	5	11	-	4	-
			4.5	25	44	-	20	-
			6.0	29	52	-	24	-
Minimum Pulse Width (CK, CK INH)	t _{w(L)} t _{w(H)}		2.0	-	30	75	-	95
			4.5	-	8	15	-	19
			6.0	-	7	13	-	16
Minimum Pulse Width (S/ $\bar{L}$)	t _{w(L)}		2.0	-	30	75	-	95
			4.5	-	8	15	-	19
			6.0	-	7	13	-	16
Minimum Set-up Time (PI-S/ $\bar{L}$)	t _s		2.0	-	15	50	-	65
			4.5	-	3	10	-	13
			6.0	-	3	9	-	11
Minimum Set-up Time (SI-CK, CK INH)	t _s		2.0	-	10	50	-	65
			4.5	-	2	10	-	13
			6.0	-	2	9	-	11
Minimum Set-up Time (S/ $\bar{L}$ -CK, CK INH)	t _s		2.0	-	30	75	-	95
			4.5	-	8	15	-	19
			6.0	-	7	13	-	16
Minimum Hold Time (PI - S/ $\bar{L}$) (SI - CK, CK INH)	t _h		2.0	-	-	5	-	5
			4.5	-	-	5	-	5
			6.0	-	-	5	-	5
Minimum Hold Time (S/ $\bar{L}$ - CK, CK INH)	t _h		2.0	-	30	75	-	95
			4.5	-	8	15	-	19
			6.0	-	7	13	-	16
Minimum Removal Time (CK INH - CK) (CK - CK INH)	t _{rem}		2.0	-	30	75	-	95
			4.5	-	8	15	-	19
			6.0	-	7	13	-	16
Input Capacitance	C _{IN}			-	5	10	-	10
Power Dissipation Capacitance	C _{PD(1)}			-	95	-	-	-

Note (1): C_{PD} is defined as the value of internal equivalent capacitance of IC which is calculated from the operating current consumption without load (refer to Test Circuit). Average operating current can be obtained by the equation hereunder.

$$I_{CC(0pr.)} = C_{PD} \cdot V_{CC} \cdot f_{IN} + I_{CC}$$

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SWITCHING CHARACTERISTICS TEST WAVEFORM

 $I_{CC(Oper.)}$ TEST CIRCUIT